

PLatS: Papyrus Surface Segmentation in Latent Space

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Abstract

Separating tightly packed papyrus sheets requires preserving their geometry and identity: a small false-positive bridge can join neighboring layers even when voxel overlap is high. We present PLatS, which learns a single-sheet latent space and predicts its codes from CT and positive point prompts. A transformer trained with latent MSE uses the same representation for sheet reconstruction and automatic seed clustering, postponing dense decoding until instances are selected. Across 106 released Kaggle test cubes, 87.3% of 764 eight-point sheet predictions have one component and no tunnels or cavities before cleanup; single-point codes distinguish annotated sheet identities with 0.912 AUROC. Automatic union scores remain below the Kaggle winner. We analyze topology errors caused by ignore-mask erasure and demonstrate independently flattened, centimeter-scale CT and ink outputs. The results support a useful sheet prior while exposing incomplete coverage, instance errors and stitching drift.

1. Introduction

Virtual unwrapping needs surfaces that stay on one papyrus sheet. In densely packed CT volumes, a few erroneous voxels can connect neighboring layers; removing those voxels after binary segmentation requires recovering which sheet they belong to. Touching sheets remain a difficulty in the Kaggle winner’s account [2]. This motivates predicting individual sheet geometry together with a representation of sheet identity.

Our central idea is to use *one latent code for reconstruction and instance discovery*. A denoising autoencoder learns a compact representation of individual sheets. Given CT and one or more points, a transformer regresses that representation with MSE. The decoded code reconstructs a sheet; distances between codes group automatically sampled foreground points. For a 320^3 crop, prompt-dependent prediction operates on only 10^3 spatial sites. Dense sheet decoding is required only for retained instance hypotheses.

We contribute (i) a point-conditioned sheet representa-

tion that couples a learned shape prior with latent identity supervision; (ii) an automatic seed–code–cluster–decode procedure that shares CT computation; and (iii) an evaluation separating raw sheet topology, automatic instance recovery, and topology changes introduced by annotation-ignore erasure. Large independently flattened surfaces expose downstream coverage and drift. Topology preservation is empirical: neither the architecture nor its losses guarantee it, and automatic segmentation does not yet exceed the winner’s combined score.

Relation to prior work. ACNN learns anatomical representations to regularize image analysis [3]; GSPN reconstructs shape proposals for point-cloud instances [8]. PointSAM studies promptable 3D point-cloud segmentation [9]. PLatS applies a shared reconstruction and identity code to thin volumetric sheets, with clustering before dense decoding. Betti matching provides spatially matched topology losses and metrics [6]; here it is used for evaluation, while the sheet prior is learned without an explicit topology loss. These connections motivate the design; they are not matched baseline comparisons.

2. Method

PLatS first learns a single-sheet code, then predicts it from CT and points. At inference, the same code space supports prompted reconstruction and automatic instance discovery. Detailed modules and losses are in Supplementary.

2.1. Learn a sheet representation

An autoencoder learns to reconstruct a clean sheet mask S from its corrupted version $\tilde{S}$:

$$z = E(\tilde{S}), \quad Q = \text{sigmoid}(D(z)). \quad (1)$$

E is the sheet encoder, z its latent code, D the occupancy decoder, and Q the reconstructed probability map; sigmoid is the sigmoid function. There are no encoder–decoder skip connections: the shape must pass through z (Fig. 1). Occupancy heads use BCE and Dice losses; dense and point-query distance heads use Smooth L1 supervision. Different-

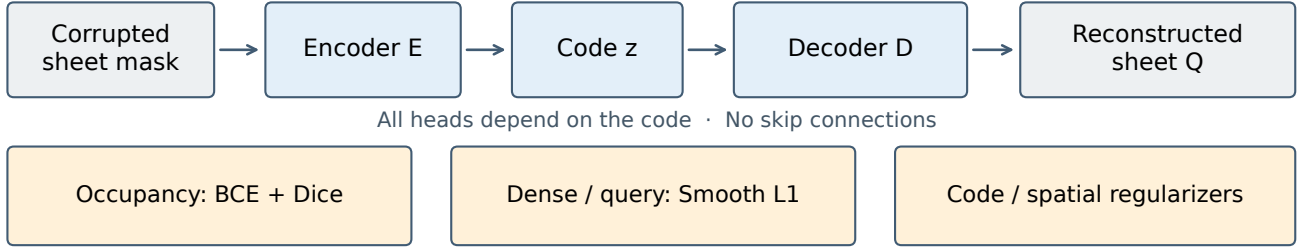


Figure 1. **Learn the sheet space.** Reconstruct a clean sheet from a corrupted mask through a latent bottleneck. Auxiliary distance heads and regularizers shape the representation. After training, freeze the AE.

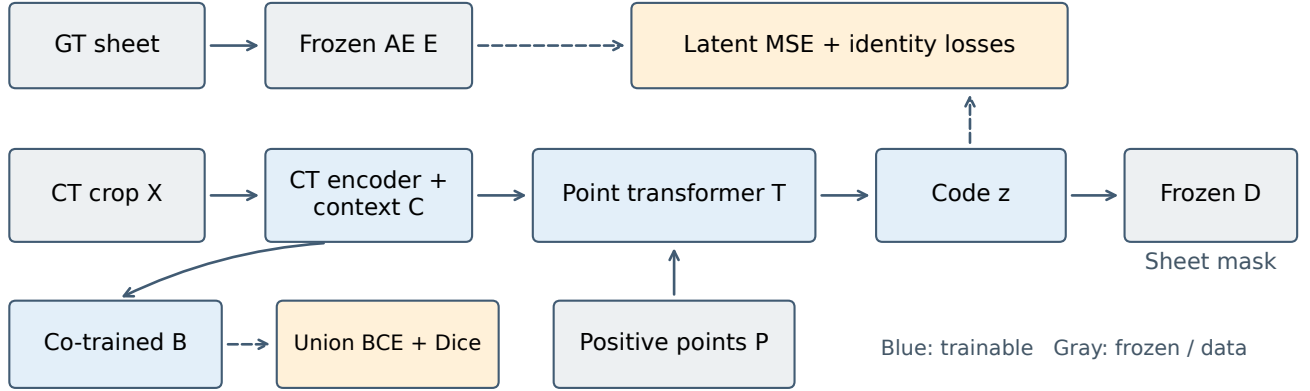


Figure 2. **Learn point-to-sheet prediction.** CT and positive points predict the frozen AE’s sheet code. The binary branch jointly supervises CT features; the new automatic recipe also uses it for seed proposal. Blue modules are trained; gray modules are frozen or inputs. The frozen sheet decoder produces the final mask at inference.

sheet code repulsion and outside-sheet/border penalties regularize training. Exact heads, noise and loss weights are in Supplementary. No explicit topology loss is used.

2.2. Predict the code from CT and points

CT encoding and image-context blocks C extract a shared feature grid. A transformer T combines these features with a set of positive points P to predict a sheet code:

$$\hat{z} = T(C(X), P), \quad \hat{S} = \mathbf{1}[\text{sigmoid}(D(\hat{z})) \geq 0.5]. \quad (2)$$

X is the CT crop, $\hat{z}$ the predicted code, $\hat{S}$ the binary sheet prediction, and $\mathbf{1}$ the indicator function. We train code prediction with MSE against the frozen AE’s clean-sheet code; normalization is omitted here for clarity and defined in Supplementary. Same-sheet and different-sheet losses encourage consistent instance identity. A co-trained binary decoder B shares the CT encoder and image context, then predicts the sheet union with BCE and Dice supervision (Fig. 2). The new training recipe crops only before the final upsampling stage, retaining earlier full-volume decoder context. The AE remains frozen, and the main point-to-sheet model needs no decoded occupancy loss during train-

ing. Prompt conditioning and latent prediction operate on a 10^3 grid for a 320^3 crop; full sheet decoding is deferred until a mask is needed. Supplementary details the modules, supervision and computational scope.

2.3. Discover instances automatically

Encode CT once and use the co-trained binary decoder to propose foreground seeds. Reuse the cached CT context to predict one code per seed, then cluster nearby codes using full-code MSE. For each retained cluster, combine consistent multi-point predictions and decode a sheet. Remove small components and duplicates, then assign overlapping voxels by predicted probability. Figure 3 shows an actual example. The reported automatic benchmark fixes a separate foreground proposer and 512 seeds per crop to control inputs across checkpoints; its measured outputs are retained in Fig. 3. The co-trained proposal path is the new recipe and has not been benchmarked here. Thresholds and exclusions are in Supplementary. Raw prompted topology is measured before cleanup.

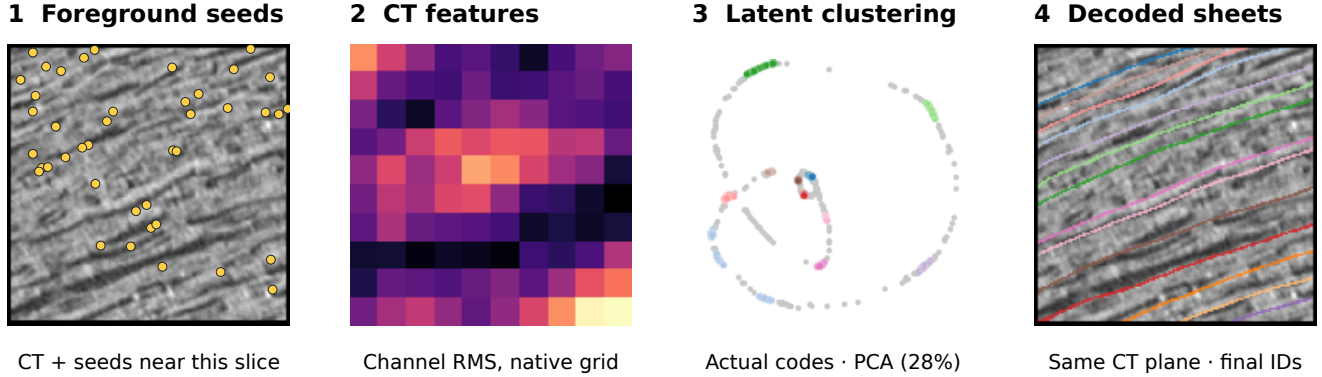


Figure 3. **Automatic inference on a real CT crop.** Left to right: foreground seeds, cached CT features, predicted codes grouped by identity, and decoded instances. The CT and output use the same native z slice of case 00860. The code plot is a 2D PCA display of all 512 measured codes; clustering uses their full dimension. Gray points are discarded. Colors link retained codes to final sheet IDs. See Supplementary for display details.

Method	Dice	SD ₂	Topo	VOI	LB
Winner	0.6015	0.8799	0.3769	0.5616	0.6176
PLatS	0.4262	0.7494	0.2644	0.5396	0.5305

Table 1. Original source-ignore erasure on all 106 released test cubes. Equal-case means; automatic inference, no GT prompts. LB is the public formula reproduced locally.

Method	Dice	SD ₂	Topo	VOI	LB
Winner	0.5460	0.8078	0.5694	0.5719	0.6537
PLatS	0.4046	0.7120	0.5820	0.5390	0.6124

Table 2. Restoration diagnostic on the same 106 cases, using the full-size 5,000-voxel gate. All three terms use restored predictions and erased GT. This is not official LB.

3. Experiments

Setup. We use Kaggle-only checkpoint **0058** for the main comparisons and evaluate all **106 released test cubes**. Training-case IDs are excluded, but labels are now public and some crops were previously inspected; this is not a fresh blind submission. External-data checkpoints, training details and all 1/2/4/8-point results are in Supplementary. We evaluate automatic union quality, ignore-erasure sensitivity, prompted geometry and identity, then downstream surfaces. The new scratch recipe has no accuracy results yet.

3.1. Binary segmentation and topology

We first apply the original leaderboard protocol: erase source-ignore voxels from prediction and GT, then compute the public formula [7]:

$$J = 0.35 \text{SD}_2 + 0.30 \text{Topo} + 0.35 \text{VOI}. \quad (3)$$

J is the combined score, SD_2 Surface Dice at two voxels, Topo exact matched TopoScore [6], and VOI the public variation-of-information score. Table 1 compares the released winner with automatic PLatS, using no GT prompts and equal weight per cube.

The winner achieves the higher combined score (0.6176 versus 0.5305) and higher Dice. Its binary mask can still connect distinct sheets (Fig. 4, top). PLatS separates layers

in this example while missing bands. Thus the example illustrates a topology failure mode, while the full benchmark identifies coverage and geometric accuracy as limitations.

3.2. Separating ignore cuts from model defects

Erasing ignored labels can create holes in an otherwise continuous prediction (Fig. 4, bottom). We therefore also report an *ignore-restoration diagnostic*: restore cut instances from their own pre-erasure predictions, exclude entirely ignored instances, and remove instances smaller than 5,000 voxels *after restoration*. The winner’s identities are raw connected components; PLatS provides sheet IDs. GT remains erased. Table 2 uses the same three formula terms. This is *not an official leaderboard score*: restored unknown foreground can still incur penalties. The exact rule and filtering controls are in Supplementary.

The winner still has the higher combined score (0.6537 versus 0.6124). PLatS has the higher restored topology term (0.5820 versus 0.5694), but lower surface and VOI scores. Restoration therefore diagnoses erasure sensitivity; it does not establish a new challenge ranking.

3.3. Sheet geometry and latent identity

To separate representation quality from automatic seed selection, we use nested GT prompts on 764 eligible annotated bands. Before cleanup or ignore erasure, 87.3% of

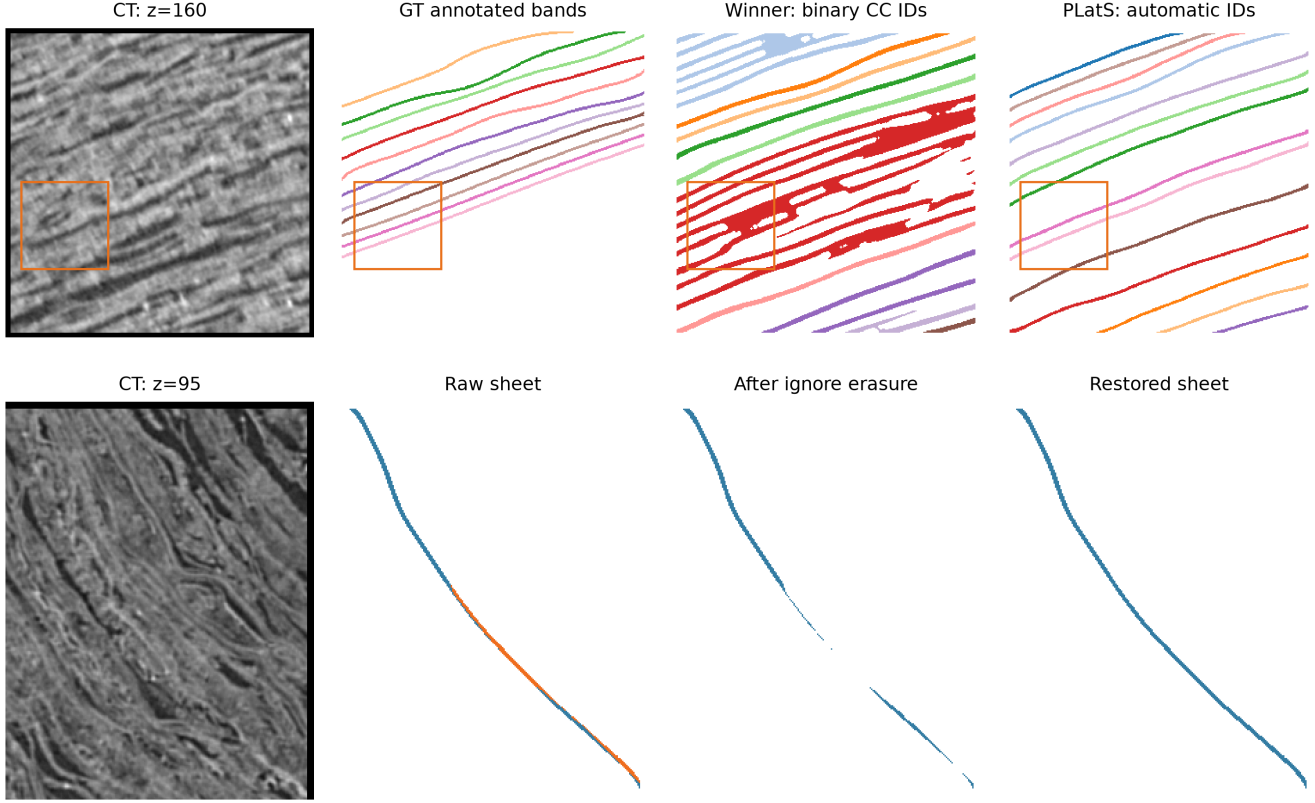


Figure 4. **Two different topology failures.** Top: a winner component joins six annotated bands in case 00860 ($z = 160$); PLaTS preserves gaps but misses bands. Colors are IDs within each method. Bottom: ignore erasure cuts a predicted sheet in case 00829 ($z = 95$). Its whole-volume topology changes from one component and no tunnels to 495 components and 1,023 tunnels. Restoring its own voxels reverses the cuts. Orange marks removed voxels. Both rows use native z slices; selection rules and the enlarged orange-square crop are in Supplementary.

Points	Dice	SD ₂	Topo	VOI	Formula
1	0.4314	0.7689	0.6864	0.7502	0.7376
8	0.4517	0.8058	0.7045	0.7440	0.7538

Table 3. Prompted 0058 sheets with no cleanup. Only the exterior of the annotated-region box is ignored; internal unannotated voxels are background. All three formula terms are shown. This policy and GT prompts differ from the automatic original-erasure comparison in Table 1.

eight-point predictions have one component and no tunnels or cavities. This is a topology statistic, not a geometric correctness rate. Table 3 measures alignment: increasing prompts from one to eight raises Surface Dice from 0.7689 to 0.8058, while voxel Dice remains modest. GT components and repaired training targets favor simple sheet topology.

For 6,112 single-point codes, same-band versus different-band classification has 0.9121 AUROC. The

distributions overlap: at the inherited MSE threshold 0.02, 50.03% of same-band pairs exceed it. Automatic instance accuracy, $TP/(TP+FP+FN)$, is 0.420 at IoU 0.25 but only 0.071 at IoU 0.50. Useful code separation therefore coexists with clustering and alignment errors. Supplementary reports complete prompt counts, instance diagnostics and raw topology; none is a causal ablation of the AE or identity losses.

3.4. From sheets to ink

Predictions are meshed, independently flattened, and passed to a frozen ink model [1, 4, 5]. Figure 5 shows two PHerc0139 regions of approximately 20×33 and 23×44 mm. Each begins with one published-track point and grows using predicted points; reference layouts and ink labels are excluded from reconstruction. Median accepted join distances are 1.181 and 1.361 voxels, versus reference-distance p90 values of 15.856 and 15.234 voxels: local continuity does not ensure global alignment. Page A covers 17.37% of annotated pixels. Gaps and diffuse ink remain.

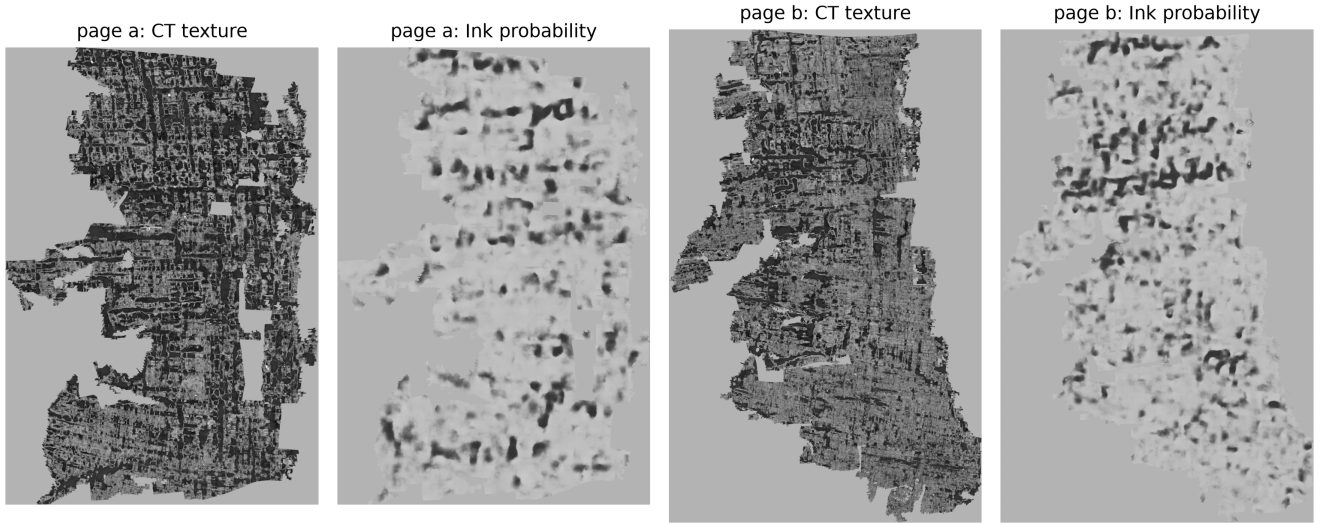


Figure 5. **Centimeter-scale surfaces and ink.** Independently flattened CT texture and ink probabilities from PLaTS on two PHerc0139 regions. Darker ink pixels indicate higher probability. Whole charts expose gaps and stitching limitations; native PNGs accompany the report.

These outputs demonstrate the pipeline, not new text recovery; the ink model’s training includes this scroll.

4. Discussion and conclusion

PLaTS uses a shared sheet code for prompted reconstruction and automatic instance discovery. Raw topology and latent separability support this design, while union scores, strict instance matching and large-page drift expose its current limits. Matched direct-decoder and loss ablations are still needed to isolate the contribution of the bottleneck and identity supervision. Measured GPU time and memory are needed to quantify the benefit of clustering before decoding. Spatially disjoint scroll evaluation and broader delivered ink coverage are needed to establish generalization and reading benefit.

References

- [1] Hob3rMallow. ScrollFiesta: Virtual meshing and unwrapping for the herculaneum papyri. https://github.com/Hob3rMallow/scrollfiesta_public. Local extraction driver built from revision 5d957e9c21529580cc43eda8bf980fef02486b36.
- [2] Tony Li, Ozan M., Yiheng Wang, and Paul G. 1st place solution for the vesuvius challenge – surface detection competition. Kaggle solution writeup, February 27, 2026, 2026.
- [3] Ozan Oktay, Enzo Ferrante, Konstantinos Kamnitsas, Matthias Heinrich, Wenjia Bai, Jose Caballero, Stuart Cook, Antonio de Marvao, Timothy Dawes, Declan O'Regan, Bernhard Kainz, Ben Glocker, and Daniel Rueckert. Anatomically constrained neural networks (ACNN): Application to cardiac image enhancement and segmentation. *arXiv preprint arXiv:1705.08302*, 2017.
- [4] Michael Rabinovich, Roi Poranne, Daniele Panozzo, and Olga Sorkine-Hornung. Scalable locally injective mappings. *ACM Transactions on Graphics*, 2017.
- [5] ScrollPrize. 9- μ m ink detection model. https://huggingface.co/scrollprize/ink_9um. Hybrid 3D/2D checkpoint seed42, step 75000, revision 7109667e2607db1b90c37c8b09cb876ea7fe7bb1.
- [6] Nico Stucki, Johannes C. Paetzold, Suprosanna Shit, Björn Menze, and Ulrich Bauer. Topologically faithful image segmentation via induced matching of persistence barcodes. *arXiv preprint arXiv:2211.15272*, 2022.
- [7] Vesuvius Challenge. Surface detection: Evaluation. Kaggle competition and public metric resource package, 2026.
- [8] Li Yi, Wang Zhao, He Wang, Minhyuk Sung, and Leonidas Guibas. GSPN: Generative shape proposal network for 3D instance segmentation in point cloud. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019.
- [9] Yuchen Zhou, Jiayuan Gu, Tung Yen Chiang, Fanbo Xiang, and Hao Su. Point-SAM: Promptable 3D segmentation model for point clouds. *arXiv preprint arXiv:2406.17741*, 2024.

Supplementary Material

A. Exact objectives and implementation

A.1. Autoencoder and full training objectives

Let X be a CT crop of $H \times W \times N_z$ voxels, S_i its binary mask for sheet i , and $P_i = \{p_k\}_{k=1}^K$ its K positive prompt coordinates. Here H, W, N_z are the three spatial dimensions. The AE encoder E maps a corrupted mask $\tilde{S}_i$ to a code z_i ; its occupancy decoder D predicts probability Q_i :

$$z_i = E(\tilde{S}_i), \quad Q_i = \text{sigmoid}(D(z_i \odot (1 + \xi))). \quad (4)$$

Here sigmoid is the logistic function, $\odot$ is elementwise multiplication, and ξ is independent uniform noise in $[-0.02, 0.02]$ during AE training. At 320^3 input, the code is 64×10^3 . No encoder–decoder skip connections bypass it (Fig. 1). The other heads predict a dense unsigned distance field U_i , a distance $h(z_i, q)$ at query coordinate q , and four intermediate occupancy maps $Q_i^{(a)}$.

Define $\mathcal{L}_{\text{seg}}$ as weighted BCE plus soft Dice loss. The AE uses

$$\begin{aligned} \mathcal{L}_{\text{AE}} = & \mathcal{L}_{\text{seg}}(Q_i, S_i) + \sum_a \alpha_a \mathcal{L}_{\text{seg}}(Q_i^{(a)}, S_i^{(a)}) \\ & + \lambda_U \mathcal{H}(U_i, d_i) + \lambda_q \mathbb{E}_q \mathcal{H}(h(z_i, q), d_i(q)) \\ & + \lambda_r \mathbb{E}_{i \neq j} [\text{ReLU}(\cos(z_i, z_j) - m)] \\ & + \lambda_o R_o(Q_i, S_i) + \lambda_b R_b(Q_i, S_i). \end{aligned} \quad (5)$$

The index a denotes decoder stages; $S_i^{(a)}$ is the resized target. $\mathcal{H}$ is Smooth L1 loss, d_i is unsigned distance to the sheet, clipped at 16 voxels and divided by 16, and $d_i(q)$ is its sampled value. The cosine term repels codes of distinct sheets in the same crop; $m = 0.5$ is its margin and $\text{ReLU}(t) = \max(t, 0)$. R_o and R_b penalize foreground outside a two-voxel GT tolerance and at the volume border. The α_a and λ symbols are loss weights, specified below. The sheet prior uses no explicit topology loss.

A.2. Point-conditioned model and joint binary head

The CT encoder C produces a 512×10^3 grid and four context attention blocks. Each prompt token combines Fourier coordinates, trilinearly sampled CT features and its positive label. Prefix attention and four transformer blocks T predict the normalized sheet code (Fig. 2):

$$\begin{aligned} \bar{z}_i &= (E(S_i) - \mu)/\sigma, \quad \hat{z}_i = T(C(X), P_i), \\ \hat{S}_i &= \mathbf{1}[\text{sigmoid}(D(\mu + \sigma \odot \hat{z}_i)) \geq 0.5]. \end{aligned} \quad (6)$$

μ and σ are fixed channel means and safe standard deviations of AE codes; $\mathbf{1}$ is an indicator. The AE remains frozen. A separate binary head B , with its own context refiner and convolutional upsampling, predicts sheet union probability

$V = \text{sigmoid}(B(C(X)))$ and supervises the shared CT features.

For transformer round r , let $a_i^{(r)}$ denote its auxiliary code, and let $d(u, v) = \|u - v\|_2^2 / L$ be MSE over L code elements. Training minimizes

$$\begin{aligned} \mathcal{L}_{\text{train}} = & d(\hat{z}_i, \bar{z}_i) + \sum_{r=1}^3 \beta_r d(a_i^{(r)}, \bar{z}_i) \\ & + 0.5 \mathcal{L}_{\text{same}} + 0.25 \mathcal{L}_{\text{diff}} \\ & + \mathcal{L}_{\text{seg}}^\Omega(V, Y), \end{aligned} \quad (7)$$

$$\begin{aligned} \mathcal{L}_{\text{same}} &= \mathbb{E}_{i,P} d(\hat{z}_i, P, c_i), \\ \mathcal{L}_{\text{diff}} &= \mathbb{E}_{i \neq j} [\text{ReLU}(1 - d(\hat{z}_i, \hat{z}_j))]. \end{aligned}$$

c_i is the mean code of same-sheet prompt sets P in a batch. Different-sheet pairs share a CT crop. $\beta = (0, 0.5, 1)$ weights the first three auxiliary heads. The final code uses a separate LayerNorm and linear head with weight 1; the instantiated fourth auxiliary head is unused by this loss configuration. Y is the GT binary union and Ω its non-ignored voxels. The binary loss uses positive BCE weight 2 and soft Dice on 160^3 crops. Point-to-sheet training uses no decoded occupancy loss in the main model.

A.3. Connections between the AE modules

Figure 6 expands the AE heads and their supervision. The reference dense encoder begins with a factor-four patch stem, followed by stage-matched residual convolutions and downsampling to a 10^3 grid. The code has 64 channels. The implementation also retains an occupied-voxel sparse encoder for future efficiency experiments; it densifies its bottleneck for the same decoder and heads. It is not used by the reported checkpoints.

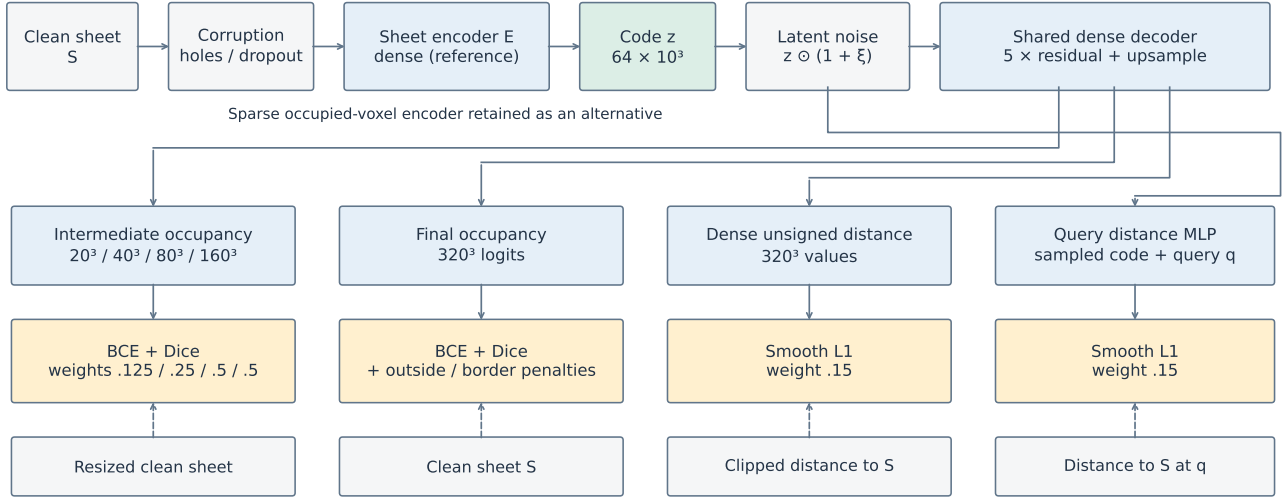
The dense decoder applies five residual/convolutional upsampling stages, with spatial sizes 10, 20, 40, 80, 160, 320 and channels 64, 48, 32, 24, 16, 8. Final $1 \times 1 \times 1$ heads predict occupancy logits and unsigned distance. Four intermediate occupancy heads attach at spatial sizes 20, 40, 80, 160. The query head instead samples the noisy code at a query coordinate, concatenates normalized coordinates, and applies a width-128 MLP. There are no encoder–decoder skip connections.

A.4. Connections between the P2SD modules

Figure 7 separates shared image computation from prompt-dependent computation. The image encoder reduces CT to a 512×10^3 feature grid. A $1 \times 1 \times 1$ projection and four self-attention blocks form the image context F : 1,000 tokens of width 512. Three-dimensional per-axis rotary coordinates supply position information.

For each prompt coordinate p_k , prompt composition concatenates its eight-band Fourier encoding, the trilinearly sampled context $F(p_k)$, and the positive-label scalar. A

AE: learn a single-sheet code and freeze it for P2SD



Code regularization: distinct sheets in one crop → cosine repulsion on z (weight .1, margin .5).

All reconstruction heads depend on the bottleneck. No encoder-decoder skips; no explicit topology loss.

Figure 6. **AE modules and supervision.** Decoder occupancy and distance heads share dense features; query distance is predicted directly from the noisy latent code. Each loss is paired with its target. Intermediate heads attach at successive decoder stages. Code repulsion acts on distinct-sheet codes before latent noise. The sparse encoder remains an optional alternative.

two-layer MLP produces one width-512 token per point. The reference retains all K prompt tokens. The prompt modulator prefixes them to the context tokens, aligns their coordinate frames for rotary attention, and applies one attention block. Prompt-prefix outputs are discarded; the remaining 1,000 conditioned grid tokens enter the refiner.

The refiner contains four self-attention blocks with eight heads and SwiGLU MLPs. Linear projections supply intermediate code estimates; a separate LayerNorm and linear projection supply the final normalized 64-channel code. The loss distinguishes these auxiliary projections from the final head, as specified in Eq. 7. Channel statistics come from training-sheet AE codes. Inference reverses this normalization before the frozen AE decoder.

The binary branch receives the *unprompted* image context, applies its own four attention blocks, and up-samples with residual/convolutional stages of channels 512, 256, 128, 64, 32, 16. A final $1 \times 1 \times 1$ head produces union logits. It has no encoder-decoder skip connections; residual connections inside individual blocks are distinct. Its loss updates the shared image encoder and image-context blocks. In the new training recipe, refinement sees the complete 10^3 context and decoding builds the full 160^3 intermediate feature grid. We crop 80^3 features there and apply the final $\times 2$ upsampling, residual block and occupancy head to obtain 160^3 output supervision. Earlier stages retain full-volume context at increased activation-memory

cost. This is the retained `second_last` crop mode, also used by 0076. The scored 0058 checkpoint instead used an earlier 5^3 context crop; its measurements are unchanged.

A.5. Efficiency of compact latent prediction

For a 320^3 crop, prompt conditioning and code prediction operate on only $10^3 = 1,000$ spatial sites, compared with 32,768,000 output voxels. Including its 64 channels, a sheet code has 64,000 scalars, 512 times fewer than one full-resolution scalar mask. At BF16, these tensors occupy 0.122 MiB and 62.5 MiB, respectively; the shared 512×10^3 context occupies 0.977 MiB. These are tensor-size comparisons, not measured speedups or total training-memory estimates. Attention still incurs token-pair work.

Let N be the number of foreground seeds, G the number of additional multi-point prompt groups used for cluster consistency, and K the number of sheet hypotheses that are densely decoded. Let C_{img} denote shared CT encoding/context cost, C_{code} the cost per prompt group, C_{cluster} clustering cost, C_B the binary branch’s refinement and union decoding cost, and C_D one AE decoding cost. With the co-trained proposal head, automatic inference has approximate cost

$$C_{\text{img}} + C_B + (N + G)C_{\text{code}} + C_{\text{cluster}} + KC_D. \quad (8)$$

This avoids paying a full upsampling cost for each of the N seed codes before clustering. Retained sheets still require

P2SD: new training recipe with late-stage binary cropping

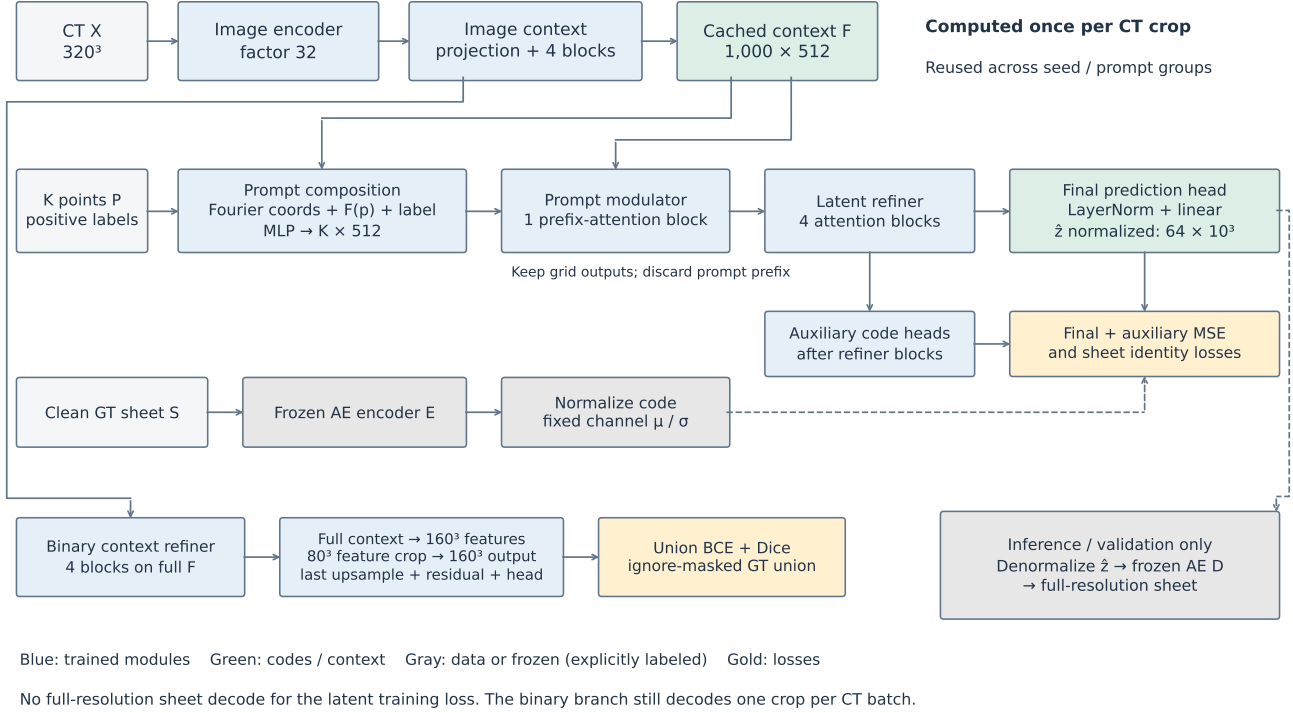


Figure 7. **P2SD modules in the new training recipe.** Shared CT encoding and context feed prompt composition, modulation and refinement. Code heads learn from the frozen AE. The binary branch decodes full context to 160³ features before cropping for its final upsampling; at inference it decodes the full union for seed proposal. The sheet decoder is used only when masks are needed under the main latent objective. Optional query losses are described separately; the scored 0058 checkpoint used earlier cropping.

dense decoding, output storage and cleanup. The co-trained proposal head reuses the cached image context and incurs one full-resolution union decode. The historical benchmark instead uses a separate proposer with its own CT encoder, so its cost includes an additional encoding. No binary training crop is applied at inference.

During P2SD training, the frozen teacher encoder produces sheet targets, but the active latent objective does not require AE decoding. The auxiliary binary branch still decodes a crop once per CT batch, and validation decodes full sheet masks. We therefore claim an architectural advantage in repeated prompt processing, rather than an unmeasured end-to-end runtime advantage.

A.6. Retained query supervision

An optional P2SD-owned distance head h_θ supplies geometric supervision without full sheet decoding. It concatenates a trilinear sample of the raw predicted code $\hat{z}_i = \mu + \sigma \odot \hat{z}_i$ with normalized query coordinates q , and ap-

plies a width-128 MLP. Its direct GT loss is

$$\mathcal{L}_{\text{query}}^{\text{GT}} = \mathbb{E}_q \mathcal{H}(h_\theta(\hat{z}_i, q), d_i(q)). \quad (9)$$

Here θ denotes the trainable query-head parameters, $\mathcal{H}$ is Smooth L1, and $d_i(q)$ is the clipped, normalized GT distance defined above. The optional term updates both the head and the predicted code. Direct GT targets avoid inheriting errors from the AE query head. A separate retained distillation option uses frozen-AE query predictions as targets instead.

The P2SD-owned head currently predicts distance only. Query occupancy/distance heads and GT-target utilities remain available in the implicit AE path; a separate GT-supervised P2SD occupancy-query head is not wired in this release. These optional paths are preserved for experiments, not included in the 0058 objective or enabled by the new default scratch recipe.

A.7. AE flip/rotation consistency: proposed extension

The same geometric idea used in the P2SD continuation can be applied to the AE, but this extension is *not implemented or evaluated here*. Define $\mathcal{A} = \text{sigmoid} \circ D \circ E$ with latent noise disabled during the consistency passes. For an exact flip or right-angle rotation g , a candidate output-space objective is

$$\mathcal{L}_{\text{AE-eq}} = \text{MSE}(\mathcal{A}(g\tilde{S}), g\mathcal{A}(\tilde{S})). \quad (10)$$

$\tilde{S}$ is one corrupted sheet; g transforms both the input and output voxel grids, and MSE averages over output voxels. Both branches use the same transformed corruption and retain supervision by the corresponding clean GT masks. A small weight after reconstruction stabilizes, applied periodically, would limit the additional decoder cost. This does not assume equivariance of the latent channels. A retrained AE would require fresh latent statistics and P2SD training against its updated codes.

A.8. Loss definitions and architecture settings

For a probability map Q , binary target S , valid voxel set Ω and positive BCE weight ω , the segmentation loss in the main text is

$$\begin{aligned} \mathcal{L}_{\text{seg}}^\Omega &= \mathcal{L}_{\text{BCE}}^\Omega + \mathcal{L}_{\text{Dice}}^\Omega, \\ \mathcal{L}_{\text{BCE}}^\Omega &= -\frac{1}{|\Omega|} \sum_{v \in \Omega} [\omega S_v \log Q_v + (1 - S_v) \log(1 - Q_v)], \\ \mathcal{L}_{\text{Dice}}^\Omega &= 1 - \frac{2 \sum_{v \in \Omega} Q_v S_v + \epsilon}{\sum_{v \in \Omega} Q_v + \sum_{v \in \Omega} S_v + \epsilon}. \end{aligned} \quad (11)$$

$Q_v = Q(v)$ and $S_v = S(v)$; v indexes voxels and $\epsilon = 10^{-6}$ stabilizes the Dice ratio. Without the superscript, all voxels are valid. AE BCE uses $\omega = \min(20, \max(1, N_- / \max(N_+, 1)))$, where N_+ and N_- count foreground/background voxels in the training batch. Intermediate targets use max pooling to the head resolution. The joint binary head uses $\omega = 2$; ignore is removed from both Dice sums, and empty-target crops contribute BCE but are excluded from the batch Dice average.

In Eq. 5, the four intermediate weights are $\alpha = (0.125, 0.25, 0.5, 0.5)$ and $(\lambda_U, \lambda_q, \lambda_r, \lambda_o, \lambda_b) = (0.15, 0.15, 0.1, 0.05, 0.05)$. Smooth L1 uses transition width 0.1 and averages over dense voxels or query samples. The penalties are

$$R_o = \frac{\sum_{v \notin \text{dilate}_2(S)} Q(v)}{\max(|S|, 1)}, \quad R_b = \frac{\sum_{v \in \partial_5} Q(v)}{\max(|S|, 1)}. \quad (12)$$

dilate_2 is a cubic two-voxel dilation and ∂_5 is the five-voxel crop border. Cosine similarity flattens the latent tensor. AE input corruption is applied with probability 0.7: one to six

spherical holes of radius 2–10 voxels and dropout up to 0.3, retaining at least 25% of the sheet. The unsigned-distance query head uses 8,192 queries.

The AE uses convolutional channels (8, 16, 24, 32, 48, 64), RMS normalization, and factor-32 reduction; its distance-query MLP has width 128. The CT encoder channels are (16, 32, 96, 192, 384, 512), with an early factor-four stem. Point Fourier encoding has eight bands. Context and point-refinement attention each have four blocks, 512-dimensional features and per-axis rotary coordinates; refinement uses eight heads. The joint binary branch has its own four-block refiner and convolutional decoder channels (512, 256, 128, 64, 32, 16). Historical 0058 binary training decodes 5^3 context crops into 160^3 masks; 0076 and the new scratch recipe use the late feature crop in Fig. 7. Prompted sheet codes retain the full 320^3 crop extent. Checkpoint, configuration and latent-statistics hashes accompany the result snapshot. The AE and main CT model use Kaggle-derived t6 training labels with topology repair/cleanup; external scroll labels are excluded from the main model. Evaluation uses the released test annotations, not those repaired training labels. Training-ID exclusion alone does not certify spatial disjointness. Cached CT inputs share five-voxel image-border zeroing; GT borders are retained.

B. Automatic clustering and instance diagnostics

Figure 3: measured inference intermediates. We reuse case 00860 and its central native $z = 160$ section, as in the binary failure example. The frozen automatic run provides all 512 seed coordinates, foreground and final instances. A forward replay with the same 0058 checkpoint, BF16 precision and seed batch size 32 captures the CT context and all 64,000-dimensional point codes. Replayed retained-cluster sizes match the saved inference log. Colors identify final instances after duplicate handling, so multiple retained clusters may share a color. Gray codes belong to discarded clusters; their exclusion is not hidden from the plot. CT seed markers use the $z \pm 12$ slab for visibility, whereas the CT and mask images use the exact central plane. The feature panel shows channel RMS of the cached 10^3 CT context at the corresponding depth, without interpolation. The code panel uses deterministic PCA over all 512 codes; its two components explain 27.8% of variance. This projection is for display only: the algorithm clusters in full code space. It need not visibly separate all sheet identities. Native arrays, point codes, assignments and source hashes accompany the figure.

Measured latent identity separation. Each of the eight fixed GT coordinates per band is queried separately, yield-

ing 6,112 single-point codes across all 106 cubes. Distance uses the complete normalized code, without decoding or threshold calibration. GT connected bands provide analysis identities, rather than physical-sheet guarantees. On 106 cases and 6,112 fixed GT points, the median same-band MSE is 0.0201, versus 2.1280 between bands. Pair AUROC is 0.9121; mean per-case nearest-band accuracy is 80.80%. At the inherited 0.02 threshold, 50.03% of same-band pairs exceed the threshold and 1.31% of different-band pairs fall below it. The same-band p90 is 1.682, so the tails overlap substantially despite separated medians. These spatially spread GT points include difficult crop-edge locations and differ from the automatic foreground-seed distribution. Pairwise rejection is not the fraction of points excluded by leader clustering; nearest-neighbor links and cluster means affect assignment. Results support useful separation while identifying remaining point-conditioning errors.

The controlled foreground proposer is frozen model 0022, shared across all checkpoints; it is separate from their jointly trained binary heads. Foreground probability threshold is 0.6, with 512 uniformly sampled voxels. One cached CT context serves all single-point latent predictions. Sequential leader clustering joins the nearest cluster mean at latent $\text{MSE} \leq 0.02$; otherwise it creates a new cluster. Clusters with eight or fewer points are discarded. For seven eight-point subset predictions u_t , define

$$r_t = \min_{s \neq t} d(u_t, u_s), \quad r_{\min} = \min_t r_t. \quad (13)$$

Here $s, t \in \{1, \dots, 7\}$ index subsets, and d is normalized latent MSE as in the main text. Candidates with $r_{\min} > 0.008$ are discarded; otherwise all seven codes are averaged and decoded once at threshold 0.5. The rule checks subset agreement, not certified physical identity.

Each automatic decode retains its largest six-connected component. Three-voxel dilation with overlap-cover threshold 0.2 suppresses duplicate masks; predicted probability assigns contested voxels, and final IDs below 500 voxels are removed. Thus the automatic benchmark includes cleanup, whereas raw prompted topology does not. Both small and unstable point clusters are excluded from reconstruction. Leader order and fixed latent thresholds remain limitations.

Tables 4 and 5 retain all automatic checkpoint results on the 106 cases. The unfiltered common-foreground CC6 baseline is not a matched-cleanup architecture ablation. Instance accuracy at IoU threshold τ is $\text{TP}/(\text{TP}+\text{FP}+\text{FN})$, using greedy one-to-one annotated-band matching; TP, FP and FN are matched predictions, unmatched predictions and missed bands. It is not integrated AP. GT-normalized IoU sums matches at 0.5 and assigns zero to unmatched GT bands. Merges count predicted majority-overlap claimants for multiple GT bands; splits count GT bands receiving multiple majority claimants. Annotation components are proxies, not certified physical-sheet identities.

C. External data and continued training

The two additional checkpoints investigate whether large-scale external data and prolonged training help generalization to new scrolls. **0076** initializes from Kaggle-only 0058 and adds mesh-derived scroll patches, totaling 10,347 spatially overlapping training patches. It adds latent coordinate-distance supervision, with query emphasis near mesh vertices. **0076+100k** continues for 100,000 steps with geometric consistency. All three retain the same frozen AE. Changed data, objectives and training duration prevent a causal attribution to any one factor.

Both mixed-data checkpoints enable the P2SD-owned GT-distance term in Eq. 9, with weight 0.15, 4,096 sampled queries and additional mesh-vertex distance targets. Their frozen-AE query-distillation option is disabled. The continuation additionally compares aligned decoded probabilities under flips and right-angle rotations:

$$\mathcal{L}_{\text{geo}} = 1 - \frac{2\langle A_g, B_g \rangle + 1}{\|A_g\|_2^2 + \|B_g\|_2^2 + 1}. \quad (14)$$

A_g and B_g are original/transformed 160^3 probability blocks after inverse alignment; inner products and norms sum voxels. A nonidentity transform is sampled each step from the 16-element in-plane dihedral and axial-flip group. Corresponding 5^3 code crops are decoded, with gradients through both branches and the frozen decoder. The consistency weight is 0.5. AdamW uses stem, encoder, dense-head and other learning rates $10^{-6}, 3 \times 10^{-6}, 10^{-5}, 5 \times 10^{-6}$, weight decay 0.02, 1,000 warmup steps and cosine decay to 10% of their initial rates. These settings allow intermediate checkpoint evaluation without relying on a near-zero terminal learning rate.

D. Full prompt-count and topology results

Of the 106 released test cubes, 102 are 320^3 and four are 256^3 . Training-case IDs are disjoint; this does not establish spatial disjointness within a scroll. The first-place baseline uses its released four-model ensemble and postprocessing unchanged; only GPU routing is adapted. Across 764 raw eight-point sheet predictions, mean (b_0, b_1, b_2) is (1.183, 0.165, 0). After source-ignore erasure, mean b_0 rises to 22.857 and mean b_1 to 32.191. These whole-mask counts differ from matched TopoScore. Here b_0, b_1, b_2 count components, tunnels and cavities under six-connected foreground. In the selected binary failure crop, the raw winner union has 125 tunnels and the automatic PLatS union has three; this example is diagnostic, not a representative estimate.

Eight spatially spread positive GT points are fixed per eligible annotated band, with nested prefixes $K \in \{1, 2, 4, 8\}$ identical across checkpoints. The complete matrix has

Arm	Dice	Surface Dice	TopoScore	VOI	Formula
Foreground CC6	0.4191	0.7377	0.0883	0.5358	0.4722
0058	0.4262	0.7494	0.2644	0.5396	0.5305
0076	0.3847	0.6863	0.2269	0.5426	0.4982
0076+100k	0.3981	0.7098	0.2386	0.5403	0.5091

Table 4. Completed automatic segmentation on 106 cubes under original source ignore. One common foreground and the same 512 sampled points are used for every P2SD checkpoint. The CC6 baseline labels the foreground directly.

Arm	Acc. _{.25}	Acc. _{.50}	GT IoU. _{.50}	Pred	GT	Merges	Splits
Foreground CC6	0.031	0.007	0.038	299.953	7.642	1.189	0.000
0058	0.420	0.071	0.057	6.821	7.642	0.330	0.000
0076	0.386	0.064	0.048	6.217	7.642	0.302	0.000
0076+100k	0.385	0.059	0.046	6.396	7.642	0.321	0.000

Table 5. Per-case means of annotated-band instance diagnostics. Acc_τ is $\text{TP}/(\text{TP}+\text{FP}+\text{FN})$ at IoU τ , not integrated AP. GT IoU sums IoUs of matches at threshold 0.50 and divides by the GT count, assigning zero to unmatched bands. Pred/GT are counts; merge and split definitions are those of the inherited matching implementation.

$3 \times 4 \times 764 = 9,168$ sheet tuples and 1,272 checkpoint/prompt/case unions. Smaller native cubes are centered with 32-voxel padding; masks, points and scores are cropped back. Every tuple has aligned image, prompt, GT, prediction and ignore NIFTIs.

The prompted diagnostic ignores only the exterior of the rectangular annotated-region box, retaining source-ignore regions *inside* that box as GT background. This avoids inward erasure cuts but is distinct from the standard LB and restoration tables in the main text. Raw masks receive no cleanup. Union GT includes annotated bands below the 10,000-voxel prompt eligibility threshold. Tables 6 and 7 retain Dice and all three formula terms for every prompt count.

Table 8 counts raw topology before erasure. All three median per-sheet vectors are $(1, 0, 0)$, while mean union tunnel counts expose outlier contacts between decoded sheets. For 0058 the mean raw union b_1 is 63.292 but its median is one: mean counts must not be read as typical defects per crop. The GT component construction also favors disk-like masks. P_{100} denotes fraction $(1, 0, 0)$ and is not proof of geometric correctness. Exact TopoScore uses spatial Betti matching over eight crop octants, not a comparison of whole-mask counts alone. Our compact exact implementation reduces storage without approximating the matching calculation. Formula arithmetic, 106-case coverage, native prompt membership and unchanged checkpoint hashes are verified. Paired bootstrap results average sheets within each case and resample 106 cases 2,000 times (seed 42); source files preserve all intervals. Spatial dependence and prior inspection limit their statistical interpretation.

E. Restoration controls and failure provenance

Preserve each predicted identity *before* erasure. Let A_i be its full mask, I the source-ignore mask, and $v_i = |A_i \setminus I|$ its visible size. The restored, filtered binary union is

$$A^{\text{rest}} = \bigcup_{i: v_i > 0, |A_i| \geq 5000} A_i, \quad J_{\text{rest}} = J(A^{\text{rest}}, G \setminus I). \quad (15)$$

$|\cdot|$ denotes voxel count. Entirely ignored instances are excluded; the 5,000-voxel gate uses *full size after restoration*. GT remains erased, and restored predictions are not erased a second time. For the binary winner, identities are its raw six-connected components; PLatS supplies predicted IDs. The rule uses no GT foreground to select or complete a prediction. Table 2 reports the same three formula terms after this change. This is *not official LB*: restored foreground can be penalized against unknown GT background, and raw binary components are imperfect sheet proxies. Filtering-only controls and a topology-isolation diagnostic are in Supplementary.

The filtering-only control erases ignore after applying the same full-size ID gate. The full restoration metric is Eq. 15, including any Surface Dice or VOI penalties for retained unknown foreground. To isolate the changed topology term, we additionally report

$$J_{\text{topo-only}} = 0.35 \text{SD}_2(A^{\text{rest}} \setminus I, G') + 0.30 \text{Topo}(A^{\text{rest}}, G') + 0.35 \text{VOI}(A^{\text{rest}} \setminus I, G'). \quad (16)$$

where $G' = G \setminus I$. This hybrid is a separate non-official diagnostic; it is not the modified LB table in the main paper.

Failure examples use saved masks, never synthetic surfaces. The binary example uses a visually clear lamellar CT

Checkpoint	Points	Dice	SD ₂	Topo	VOI	Formula
0058	1	0.4314	0.7689	0.6864	0.7502	0.7376
0058	2	0.4329	0.7735	0.6994	0.7507	0.7433
0058	4	0.4471	0.7936	0.6954	0.7463	0.7476
0058	8	0.4517	0.8058	0.7045	0.7440	0.7538
0076	1	0.4050	0.7445	0.6576	0.7557	0.7224
0076	2	0.4115	0.7564	0.6546	0.7530	0.7247
0076	4	0.4287	0.7820	0.6635	0.7477	0.7344
0076	8	0.4334	0.7987	0.6772	0.7450	0.7435
0076+100k	1	0.4214	0.7621	0.6729	0.7505	0.7313
0076+100k	2	0.4269	0.7718	0.6658	0.7485	0.7319
0076+100k	4	0.4413	0.7939	0.6719	0.7446	0.7400
0076+100k	8	0.4422	0.8071	0.6821	0.7430	0.7472

Table 6. Complete sheet results: all three checkpoints, all four nested prompt counts. Annotated-box ignore; no prediction cleanup. Formula scores are diagnostics under this policy.

Checkpoint	Points	Dice	SD ₂	Topo	VOI	Formula
0058	1	0.4437	0.7922	0.5848	0.5336	0.6395
0058	2	0.4465	0.7978	0.6054	0.5352	0.6482
0058	4	0.4582	0.8132	0.6264	0.5343	0.6595
0058	8	0.4619	0.8221	0.6171	0.5341	0.6598
0076	1	0.4166	0.7692	0.5597	0.5296	0.6225
0076	2	0.4220	0.7787	0.5373	0.5322	0.6200
0076	4	0.4368	0.7994	0.5704	0.5335	0.6376
0076	8	0.4426	0.8122	0.5990	0.5305	0.6497
0076+100k	1	0.4313	0.7836	0.5781	0.5308	0.6335
0076+100k	2	0.4370	0.7928	0.5543	0.5342	0.6308
0076+100k	4	0.4495	0.8114	0.5725	0.5349	0.6430
0076+100k	8	0.4512	0.8210	0.6065	0.5313	0.6553

Table 7. Complete union results: all three checkpoints, all four nested prompt counts. Annotated-box ignore; no prediction cleanup. Formula scores are diagnostics under this policy.

Checkpoint	Per-sheet means				Union means		
	b_0	b_1	b_2	P_{100}	b_0	b_1	b_2
0058	1.183	0.165	0.000	87.30%	8.179	63.292	9.528
0076	1.385	0.658	0.263	81.94%	9.594	19.906	3.557
0076+100k	1.308	0.534	0.211	82.46%	8.953	40.302	5.896
GT	1.000	0.001	0.001	99.74%	7.642	0.009	0.009

Table 8. Whole-mask raw Betti counts before scoring erasure or component cleanup. Sheet statistics cover 764 eligible bands; union statistics cover 106 cases. P_{100} denotes the fraction of sheets with $(b_0, b_1, b_2) = (1, 0, 0)$. These counts do not replace spatial feature matching.

cross-section among the top 20 cached winner merger candidates. The central TIFF-stack slice is used, with a four-connected winner region joining distinct GT bands. The ignore example maximizes induced tunnel count among raw $(1, 0, 0)$ 0058 masks with visible support. Source hashes, selection metadata, native voxels and slice coordinates are saved in the failure-analysis run. Selection is diagnostic, not random or representative. Figures 4 and 8 show na-

tive 2D slices with nearest-neighbor display, using the same plane and crop across methods. TIFF arrays are $[z, y, x]$ and project arrays are $[x, y, z]$; a displayed z slice is the transpose of internal array $[:, :, z]$. CT panels are verified directly against the original TIFF pages. The ignore-cut slice maximizes disconnected visible components among continuous raw z sections with at least 1,000 foreground and 20 removed voxels. Counts in the main caption use the exact

Method	Standard LB	Filter only	Restore + filter	Topology-only diagnostic
Winner	0.6176	0.6176	0.6537	0.6753
PLatS auto	0.5305	0.5304	0.6124	0.6257
PLatS GT8 union	0.5627	0.5627	0.6598	0.6611

Table 9. Ignore-restoration controls. GT8 union uses deterministic sheet-ID ownership on overlaps and retains its original union exactly. It uses GT prompts and is not an automatic discovery result. The topology-only hybrid replaces only TopoScore.

whole-volume voxel V construction; a displayed 2D interruption alone does not establish a 3D tunnel.

F. Large pages and spiral-fitting integration

The large-page main-paper figure uses 0058 on the same two preselected PHerc0139 regions as our earlier downstream control. Native sampling is 9.362 μm ; patch size 320 and step 128 give 60% linear overlap. The first decoded prediction defines a tangent frame; later points come from accepted predictions. An overlap acceptance gate can reject ambiguous growth, so accepted join error is conditional on acceptance and is not a global geometric error. Independent SLIM retains holes in predicted chart support. The inward normal is fixed from the scroll umbilicus before ink inference; full inward/outward PNGs are preserved. Geometry and partial ink labels enter only after all reconstruction/ink hashes are frozen. The known ink-model scroll and potentially overlapping external training regions are not held-out transcription tests. On page A, the single-method supported-pixel ink AP/Dice are 0.9170/0.8294; labeled coverage is 17.37% and delivered Dice is 0.2932. These supported pixels are not the common-pixel set used in the integration comparison below.

An earlier integration pilot, using original 0076, compares the direct prediction, published m7 tracks passed to the official spiral fitter, PLatS patches alone, and tracks plus PLatS patches. All fits have the same 30,000-step budget, canonical initialization, independent flattening and frozen ink. The published tracks have *unverified Kaggle-winner provenance*; this is not the first-place model comparison. Two predicted local charts alone leave global fitting underconstrained. We retain these controls to demonstrate an integration route, not an upper bound on the published fitting method.

Ink AP and Dice in Table 11 use the same 26,296 fully supported labeled pixels across all four arms. Labeled coverage and “delivered” Dice also count missing output, assigning zero to missing predictions. Pairwise comparisons use different common pixel sets and must not be mixed with the all-arm table. Page B has no audited ink labels. High common-pixel scores on a small delivered fraction do not establish whole-page letter recovery. This pilot studies integration and drift.

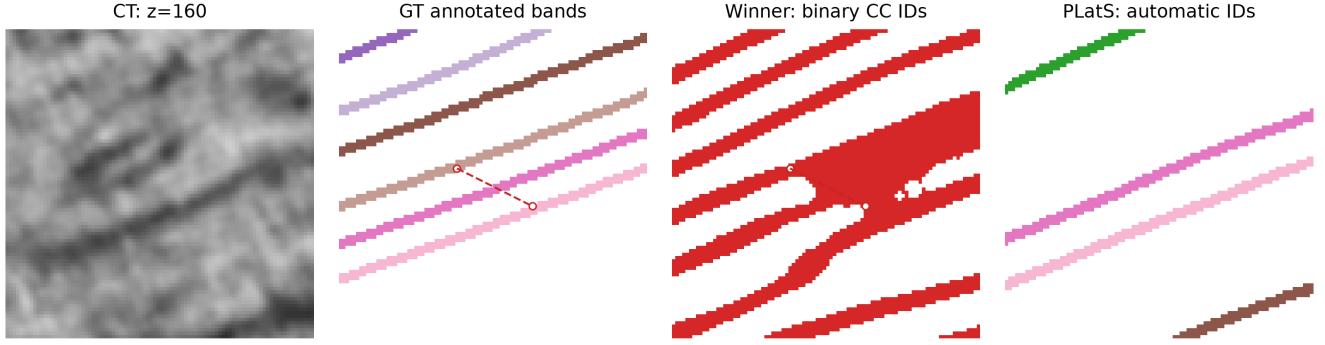


Figure 8. Enlarged native slice from the main-paper binary failure example: the same 90×90 -pixel crop at TIFF $z = 160$ in all four columns. The winner component connects distinct GT bands; automatic PLatS has lower coverage. Twenty-seven predicted voxels in the sampled gap are known background; none of its 34 samples are ignored. The red dashed segment marks this sampled gap in GT and winner panels. Labels are displayed without dilation or smoothing.

Page	Area cm^2	Accepted/attempted	Join median (vox)	Target median	Target p90
page_a	5.67	255/365	1.181	3.905	15.856
page_b	8.92	411/565	1.361	2.778	15.234

Table 10. 0058 large-page outputs. Areas are reconstructed mesh areas; joins include accepted windows only. Target distances exclude reference-boundary samples and are measured after all ink outputs are frozen.

Input/method	Area cm^2	Median vox	p90 vox	Ink AP	Ink Dice	Coverage	Delivered Dice
Direct PLatS	6.258	2.249	8.059	0.7582	0.6021	0.2348	0.3324
Published tracks	5.773	14.450	29.358	0.3314	0.1731	0.1598	0.0524
PLatS patches only	6.236	6.311	94.290	0.8393	0.7446	0.1444	0.1314
Tracks + PLatS	5.948	3.112	12.187	0.9881	0.9336	0.1591	0.2239

Table 11. Original 0076 integration pilot on page A. Common ink uses identical pixels across all four arms; coverage and delivered Dice expose missing output. Published tracks are not verified as the Kaggle winner.

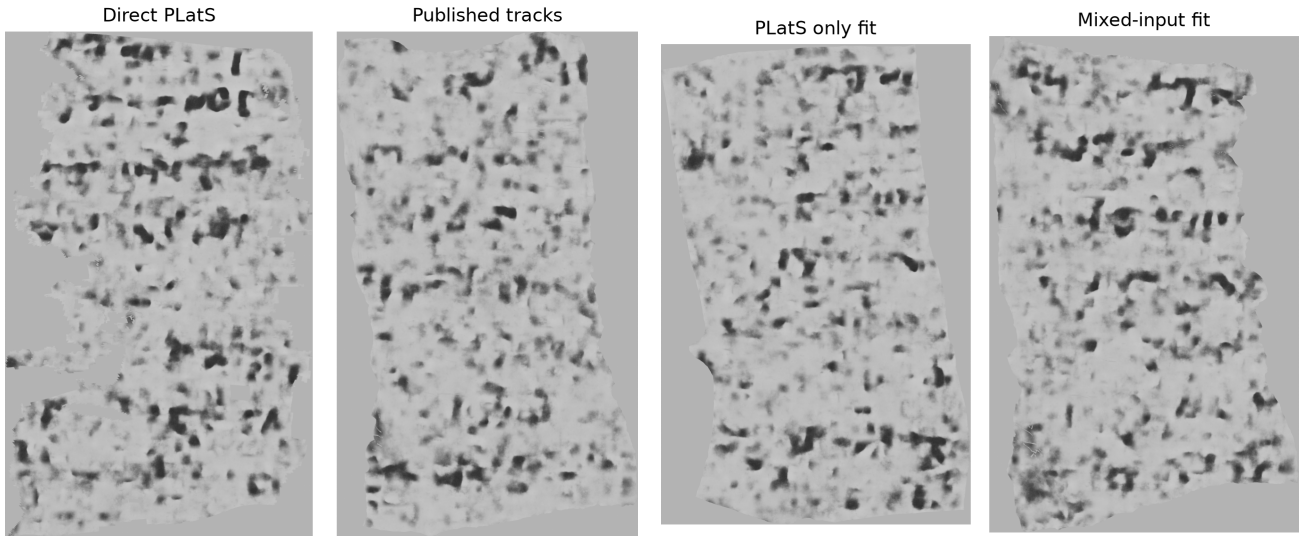


Figure 9. Large-page spiral integration control on PHerc0139 page A, original 0076 (Supplementary only). Direct prediction, published-track fitting, prediction-only fitting and mixed-input fitting all use independent layouts and the same ink model. The track source is not verified as the Kaggle winner.